



Predictors of seizure control in patients with juvenile myoclonic epilepsy (JME)



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ABSTRACT

Purpose: Juvenile myoclonic epilepsy (JME) is a widely accepted epilepsy syndrome. Factors that are potentially predictive for seizure outcome in JME are not entirely clear yet. In the current study, we tried to identify factors that are potentially predictive for seizure outcome in patients with JME.

Method: In this case–control study all patients with a clinical diagnosis of JME were recruited at the outpatient epilepsy clinic at Shiraz University of Medical Sciences from 2008 to 2012. All patients were followed for at least 18 months. We divided the patients into two groups: patients who were seizure free in the last 12 months of their follow-up period and those who had any seizures (i.e., generalized tonic-clonic (GTC), myoclonic or absences).

Results: 116 patients were studied. Sixty-eight patients were seizure free in the last 12 months of their follow-up and 48 patients had at least one seizure of any type. Shorter follow-up period, drugs other than valproate in the drug regimen, and poor drug adherence were significantly more frequent among those who were not seizure free.

Conclusion: A correct diagnosis of JME does not guarantee an optimal response to therapy. Poorer drug adherence might result in more seizures; this should be highlighted for patients in every single visit. Longer follow-up gives the treating physician more time to adjust the drug(s) and formulate a plan that is appropriate for patients with difficult to treat seizures. We recommend valproate if seizure control is the main objective.

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1. Introduction

Juvenile myoclonic epilepsy (JME) is a widely accepted syndrome of idiopathic generalized epilepsies (IGEs) with myoclonic seizures with or without other seizure types (e.g., generalized tonic-clonic seizures or absences), with age of onset often in adolescence or young adulthood and with particular EEG features including generalized spikes-waves/polyspikes and a normal background.^{1,2} Juvenile myoclonic epilepsy is one of the most common epilepsy syndromes. Frequency of JME is about 9% of all epilepsies.² Juvenile myoclonic epilepsy is often characterized by excellent response to treatment, if diagnosed correctly.³

However, drug selection and treatment plan should be individualized, considering various factors including drug efficacy, patient's gender, age, and comorbidities, among other factors. Besides, the long-term seizure outcome of JME is still controversial and factors that are potentially predictive for seizure outcome are not entirely clear yet.⁴

In the current study, we tried to identify factors that are potentially predictive for seizure outcome in patients with juvenile myoclonic epilepsy. Identification of these predictors can potentially increase the ability of physicians to manage their patients more appropriately.

2. Methods and material

In this case–control study all patients with a clinical diagnosis of JME were recruited at the outpatient epilepsy clinic at Shiraz University of Medical Sciences from September 2008 to November 2012. The diagnosis of JME was made exclusively by the only

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epileptologist working at this institution (the first author) and based on clinical grounds and electroencephalographic findings, and all patients had to be under the care of the epileptologist at our institution for at least 18 months. We considered the first six months of their follow-up as the time period to formulate an individualized treatment plan and adjust the drug regimen according to the patient's profile and response. Once JME has been diagnosed, an anti-epileptic drug (AED) appropriate for that condition may be prescribed. Our main goal of treatment was to prevent future seizures from recurring. Treatment was typically started with one of the first line AEDs. First line drugs should be effective and well tolerated. For generalized epilepsies, first line drugs include valproate, lamotrigine, levetiracetam, and topiramate among others.⁵ Our general strategy was to prescribe valproate for men and lamotrigine for women, considering other patient's profiles (e.g., comorbidity, drug history, etc.). Levetiracetam is very expensive in Iran and is not covered by many insurance policies. If a first line drug controlled seizures and did not cause side effects, therapeutic adjustments were not needed. If seizures persisted, the medication regimen was adjusted and another AED (e.g., levetiracetam, topiramate, etc.) was replaced the first AED slowly. If two monotherapies failed, we formulated a polytherapy regimen for the patient, considering the trouble making seizure type(s) and the patient's profile.

We divided the patients into two groups: patients who were seizure free in the last 12 months of their follow-up period and those who had any seizures (i.e., generalized tonic-clonic, myoclonic or absences) in that time period. Age, gender, age at seizure onset, seizure type(s) and their frequencies, epilepsy risk factors (including history of febrile seizure, positive family history of epilepsy, and parental consanguinity), social history (including tobacco and alcohol consumption) and EEG findings of all patients were registered routinely. Demographic variables and relevant clinical and EEG variables were summarized descriptively to characterize the study population. Statistical analyses were performed using Chi square, Fisher's Exact and Mann–Whitney tests to determine potentially significant differences, and a *P* value less than 0.05 was considered significant. This study was conducted with the approval by Shiraz University of Medical Sciences Review Board.

3. Results

During the study period, 116 patients with JME were eligible to enter into the study. Sixty-eight patients were seizure free in the last 12 months of their follow-up and 48 patients had at least one seizure of any type. Demographic and clinical characteristics of these two groups of patients are shown and compared in Table 1. As it is shown, in the univariate analysis, patients who were seizure free in their follow-up had longer duration of follow-up, valproate in their drug regimen and a better drug adherence, while those who experienced seizures had more frequent generalized tonic-clonic seizures (GTCS) before their first visit and more often reported alcohol and tobacco consumption. In the second step, we performed binary logistic regression on the variables with significant *P*-values in univariate analysis (mentioned above). The full model was statistically significant $X^2(6, N = 116) = 29.37, P < 0.001$, indicating that the model was able to distinguish between seizure free patients and those who experienced seizures and correctly classify them in 71.6% of cases. Among all six independent variables (*duration of follow-up, tobacco consumption, alcohol consumption, history of monthly GTCSs before the first visit, valproate in the drug regimen and good drug adherence*), only three variables made a unique statistically significant contribution to the model, as follows: good drug adherence ($P = 0.02$, OR = 3.09, 95% CI: 1.1–8.2); valproate in the drug regimen ($P = 0.02$, OR = 2.8, 95% CI: 1.1–7.6); and duration of follow-up ($P = 0.01$, OR = 1.06, 95% CI: 1.01–1.1). The median dose of valproate was 800–1000 mg per day in both groups in the follow-up visits (at 15 months, 27 months and 39 months of follow-up visits). Electroencephalographic findings of these two groups of patients are shown and compared in Table 2.

4. Discussion

Juvenile myoclonic epilepsy (JME) is a common epilepsy syndrome.² Juvenile myoclonic epilepsy is often characterized by excellent response to treatment,³ but delay to a correct diagnosis and consequently, choosing an inappropriate or suboptimal agent may result in uncontrolled seizures in many patients.⁶ Besides, even a correct diagnosis does not guarantee an optimal response to therapy and many patients suffer from ongoing

Table 1
Comparison of demographic and clinical characteristics of patients with JME, with or without seizures in their follow-up.

Variables	Seizure free patients (68 patients)	Patients with seizure(s) (48 patients)	<i>P</i> value
Sex ratio (female to male)	46/22	32/16	0.9
Age of seizure onset	15.4 ± 4.6 years	16 ± 3.2 years	0.4
Disease duration prior to the first visit	8 ± 7.1 years	10 ± 12.2 years	0.6
Duration of follow-up	37.2 ± 8.4 months	33.2 ± 10.3 months	0.02
Age at the first visit	23.1 ± 7.5 years	24.9 ± 8.8 years	0.2
Already on AED at the first visit	51 (75%)	38 (79.2%)	0.6
Tobacco consumption	2 (2.9%)	7 (14.5%)	0.02
Alcohol consumption	0	5 (10.4%)	0.01
History of monthly GTCSs before the first visit	10 (14.7%)	15 (31.2%)	0.03
History of absence seizures before the first visit	27 (39.7%)	18 (37.5%)	0.8
History of daily myoclonic seizures before the first visit	14 (20.5%)	12 (25%)	0.5
Parental consanguinity ^a	24 (35.2%)	18 (37.5%)	0.8
Family history of epilepsy (first, second, or third degree relatives)	23 (33.8%)	22 (45.8%)	0.2
History of febrile seizure	3 (4.4%)	1 (2%)	0.6
History of status epilepticus before the first visit	5 (7.3%)	0	0.07
Monotherapy regimen	52 (76.4%)	31 (64.5%)	0.16
Valproate in the drug regimen (as mono- or polytherapy)	48 (70.5%)	23 (47.9%)	0.01
Good drug adherence	58 (85.2%)	30 (62.5%)	0.005
Medical comorbidity	20 (29.4%)	21 (44.6%)	0.1
Abnormal MRI findings	4/27 (17.3%)	5/16 (31.2%)	0.3

Significant differences are in bold.

^a Consanguinity (cousin to cousin marriage) is a common practice in the Middle East.

Table 2

Comparison of the EEG findings in patients with JME, with or without seizures in their follow-up.

Variables	Seizure free patients (68 patients)	Patients with seizure(s) (48 patients)	P value
Generalized spike-waves	15 (22.1%)	17 (35.5%)	0.1
Generalized spike-waves and Polyspikes	28 (41.2%)	12 (25%)	0.07
Polyspike-waves only	9 (13.2%)	6 (12.5%)	0.9
Normal	16 (23.5%)	13 (27%)	0.6
Photoparoxysmal response	4 (5.8%)	2 (4.2%)	1
Focal abnormalities	2 (2.9%)	1 (2.1%)	1

seizures despite an appropriate treatment plan. In the current study, 58.6% of our patients with JME responded well to their treatment plan, while 41.4% experienced seizures (GTCs, absences or myoclonic jerks) in their follow up. This figure is higher than some previous reports.⁷ This discrepancy might be due to the referral nature of our clinic or alternatively, the method of the study. We considered all seizure types including rare myoclonic jerks and absences in the follow up period; while others probably considered only GTCs (74% of our patients were free of GTCs in their follow up visits).

When we compared patients who were seizure free with those who experienced any seizure(s) in their follow up in a binary logistic regression analysis, we observed that shorter follow up period; AEDs other than valproate in the drug regimen; and poor drug adherence were significantly more frequent among those who were not seizure free. Electroencephalography is a valuable ancillary test in making the diagnosis of JME.⁸ In previous studies conflicting results were observed about the role of EEG and its association with treatment response in patients with JME. In one retrospective study, focal EEG abnormalities increased the chances (though, it did not correlate with) of lack of response to valproate in these patients.⁹ In another retrospective study, EEG asymmetries were not associated with resistance to treatment in patients with JME.¹⁰ In our study, EEG could not predict seizure outcome in patients with JME. Longer follow up gives the treating physician more time to adjust the drug(s) and formulate a plan that is appropriate for patients with difficult to treat seizures. Poor drug adherence is an important factor in causing seizures in patients with epilepsy, particularly those with JME. In a study done nine years ago in the same region, drug adherence was satisfactory in 72% of the patients,¹¹ which is very similar with the current situation (75% in the current study). Drug adherence had a very significant contribution in the model that distinguished between seizure-free patients and those who experienced seizures, in the current study. This means poorer drug adherence might result in more seizures in patients with JME; something that should be highlighted for patients in every single visit. It is important to tell the patients that JME is often characterized by excellent response to treatment only if treatment is taken regularly! Superiority of valproate versus lamotrigine treatment in idiopathic generalized epilepsy syndromes including JME has been observed in many previous studies.^{12,13} We indirectly had the same observation in our study, as those taking valproate showed better outcome with regard to seizure control compared with those taking other AEDs. In patients with JME and particularly those with difficult to treat seizures, we recommend valproate, if seizure control is the main

objective. However, when pregnancy is contemplated, the patient and her family should be informed that valproate is associated with the highest risk for major congenital malformations, and it might also affect cognitive and behavioral development of offspring. This creates a significant headache when trying to choose the “right” drug for most young women with JME.¹⁴

5. Limitations of the study

- (1) This was a clinic-based series and may not represent the full spectrum of JME.
- (2) The population was heterogeneous as some were untreated (27 patients) and others have been pre-treated (89 patients). Study of the untreated patients (seizure free (17 patients) and seizure present (10 patients)) separately did not yield meaningful results, because of the small numbers.

Conflict of interest

Ali A. Asadi-Pooya has received honoraria from Cobel Daru. Zabiollah Hashemzahi and Mehrdad Emami have no conflict of interest.

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